A published-descent baseline for full-degree binary coefficients

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Classification: **derived corollary of published work**, with a Rust parameter checker. This does not establish a new vulnerable family, publication novelty, a complete attack, or a cost obstruction. It explains a baseline that a future novelty claim must distinguish from a useful low-genus or low-cost descent.

Request and current evidence

Requirement	Status
Identify a previously unrecognized affected source family or criterion	Not established. The five earlier families already have higher-genus Hess descents.
Supply a proof	The inclusion corollary below is proved from an elementary trace lemma and Hess's Theorems 11, 12 and 14. Its ingredients and construction are prior work.
Lower total cost than GHS/gGHS, or a broad cost obstruction	Not established. No construction/transfer/solve cost comparison is made here.
Rust implementation in the cryptanalysis repo	Completed by merged PR 1018. The new checker is Rust too.
Continue the pending exploration sweep	Still open: 355 denominator jobs remain. No fresh campaign data is collected in this change.
Keep prime fields and Koblitz controls in their proper scope	Relative degree one is excluded; proper-relative-subfield b, including b=1, is excluded from this corollary's assertion.

The owner's clarification remains in force: beating Pollard rho is not the acceptance gate for identifying a new affected family. Conversely, an unimplemented cost comparison is not filled in with a genus comparison.

A short trace lemma

Lemma (derived). Let $K = \mathbf{F}_{q^N}$ over $k = \mathbf{F}_q$, with $N \geq 3$, and put $U = \ker \mathrm{Tr}_{K/k}$. For every $s \neq 0$ there is a $\gamma \in U \setminus \{0\}$ such that $\mathrm{Tr}_{K/k}(s/\gamma) \neq 0$.

Proof. Otherwise all $q^{N-1} - 1$ nonzero roots of

$$L(X) = \sum_{i=0}^{N-1} X^{q^i}$$

would be roots of

$$P_s(X) = \sum_{i=0}^{N-1} s^{q^i} X^{q^{N-1}-q^i}.$$

Both P_s and L/X have degree $q^{N-1}-1$, and L/X has exactly those distinct roots. Thus $P_s = sL/X$. But the coefficient of $X^{q^{N-1}-q}$ on the left is $s^q \neq 0$. That exponent is positive and divisible by q, whereas every positive exponent in L/X is congruent to -1 modulo q. This is a contradiction. □

The $N = 2$ case is intentionally separate. In characteristic two, $U = k$; if $s \notin k$, take $\gamma = 1$, so $\mathrm{Tr}_{K/k}(s/\gamma) \neq 0$. The lemma is not asserted for arbitrary nonzero $s \in k$ in this quadratic case.

Published-construction inclusion

Corollary (derived from Hess). Let $k = \mathbf{F}_q$ have characteristic two, $K = \mathbf{F}_{q^N}$ with $N \geq 2$, and

$$E/K: \quad Y^2 + XY = X^3 + aX^2 + b, \qquad b \neq 0.$$

Suppose b has full relative degree N over k. There is a curve C_0/k obtained by Hess's generalized split-pole construction with

$$g(C_0) \leq 2^N - 2$$

and a homomorphism

$$\Phi : E(K) \longrightarrow J_{C_0}(k)$$

whose kernel is annihilated by a power of two. In particular, Φ is injective on every odd-prime-order subgroup. This applies to either absolute-trace class of a. The target need not be hyperelliptic.

Proof. Set $s = \sqrt{b}$. Choose γ by the lemma, or take $\gamma = 1$ in the quadratic case, and put $\beta = s/\gamma$. Then

$$\gamma\beta = s, \qquad \mathrm{Tr}_{K/k}(\gamma) = 0, \qquad \mathrm{Tr}_{K/k}(\beta) \neq 0.$$

The explicit substitution $X = \beta x$, $Y = \beta x w + s$ identifies the elliptic function field with

$$w^2 + w = \gamma/x + a + \beta x.$$

Write $\sigma(z) = z^q$, let $M_\gamma, M_\beta \in \mathbf{F}_2[T]$ be the minimal annihilators under σ, and let $M = \mathrm{lcm}(M_\gamma, M_\beta)$, with degrees u, v, m respectively. All annihilators divide $T^N + 1$, so $m \leq N$. The zero trace of γ gives $M_\gamma \mid (1 + T + \cdots + T^{N-1})$, hence $u \leq N - 1$. Applying relative trace to $M_\beta(\sigma)\beta = 0$ gives $M_\beta(1)\,\mathrm{Tr}_{K/k}(\beta) = 0$. Thus $T + 1$ divides M_β, and consequently M.

It follows that $\mathrm{Tr}_{K/\mathbf{F}_2}(M(\sigma)a) = M(1)\,\mathrm{Tr}_{K/\mathbf{F}_2}(a) = 0$. Hess's exact-constant condition holds for both trace classes of a, the function-field degree parameter is m, and Theorem 11 gives the Frobenius lift because γ has zero relative trace. Theorem 12 gives

$$g(C_0) = 2^m - 2^{m-u} - 2^{m-v} + 1.$$

For $m = N$, use $u \leq N - 1$ and $v \leq N$ to obtain $g(C_0) \leq 2^N - 2$. For $m < N$, the bound $g(C_0) \leq 2^m - 1 \leq 2^{N-1} - 1$ suffices.

Finally, if σ^j fixes both γ and β, it fixes $b = (\gamma\beta)^2$. Full relative degree of b therefore forces the coefficient-pair period $s(h)$ in Hess's Theorem 14 to equal N. That theorem bounds the kernel of the norm-after-conorm map by $E(K)[2^{m-1}]$. Its restriction to an odd-prime-order subgroup is injective. □

What this changes in the novelty test

For this full-relative-degree class, merely finding a prime-subgroup-preserving descent at some genus cannot establish the first connection with GHS-like descent: it is already contained in the published framework. This includes both twists, after a suitable split-pole presentation. A canonical Frobenius-lift failure does not exclude generalized descent.

The bound is an existence **upper** bound, not a lower bound or an optimal genus. It gives 2, 6, 30, 2046 and 8190 for relative degrees 2, 3, 5, 11 and 13. Those numbers do not estimate construction, transfer or solve cost. They do not rule out a new useful low-genus recognition criterion, a new explicit correspondence, or a lower-cost attack. They do not settle the publication priority of the earlier degree-eight/nine maps.

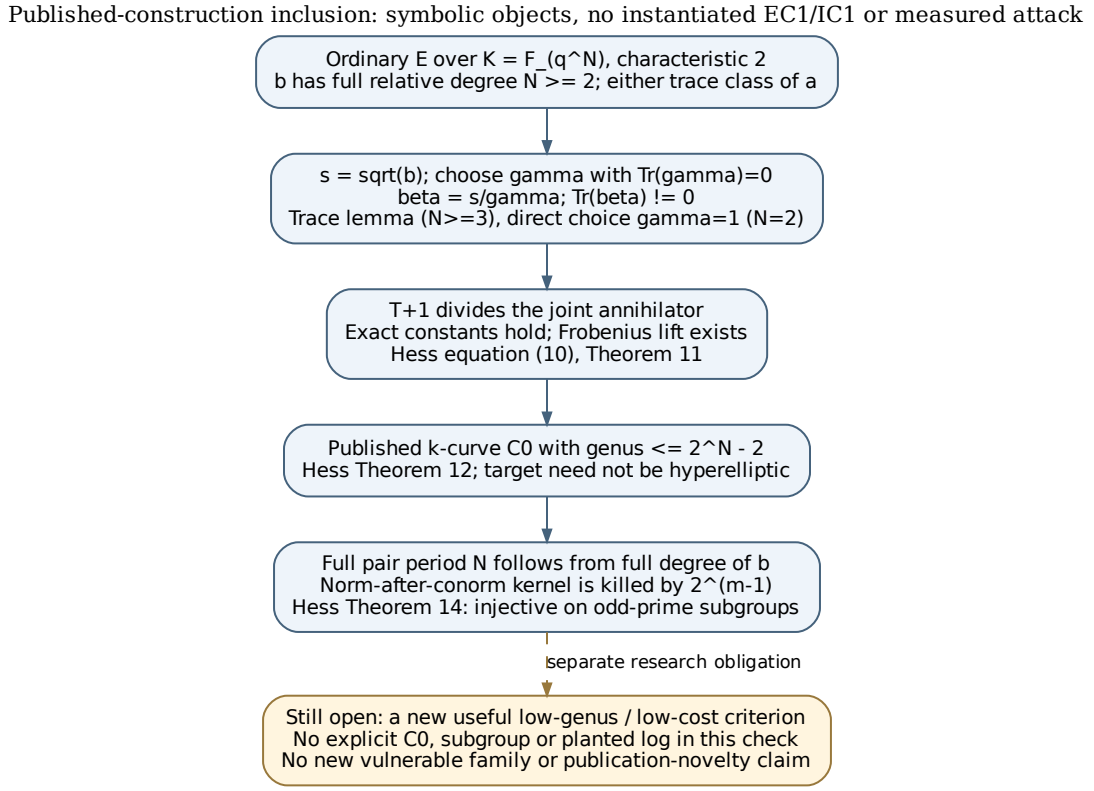


Figure 1: Symbolic inclusion and the outstanding quantitative gate. No concrete curve IDs or measured attack are represented.

The editable diagram is inclusion.dot, with an SVG rendering. It adds no curve or isogeny edge to a canonical graph.

Rust checker and evidence boundary

`examples/hess_scope.rs` constructs one (γ, β) witness for a specified full-degree coefficient. It checks the product, relative traces, annihilator relations, exact-constant condition and full joint period with the fast field and separate raw-polynomial arithmetic. It returns the genus of that published witness, not the minimum over presentations.

```
cargo run --release --locked \
  --manifest-path experiments/ghs-sweeper/Cargo.toml \
  --example hess_scope -- \
  --base-degree 1 --relative-degree 3 --a 1 --b 2
cargo test --release --locked \
  --manifest-path experiments/ghs-sweeper/Cargo.toml \
  --example hess_scope
```

The checker does not instantiate C_0, its Jacobian, a subgroup generator, or a planted log. Curve and IC candidate IDs remain null. The injectivity statement above is a proof using the published theorem, not a measured transfer claim. No catalogue entry is created.

The fixed software unit controls check the quadratic construction in both constant-trace classes, the explicit cubic witness $(\gamma, \beta) = (2, 3)$ in $\mathbf{F}_8$, an even-degree case, the historical trace-one coefficient $b = 166$ in $\mathbf{F}_{1024}$, and rejection of singular/trivial/subfield inputs. These are software correctness assertions, not a new parameter sweep, a natural-yield estimate, or an IC benchmark.

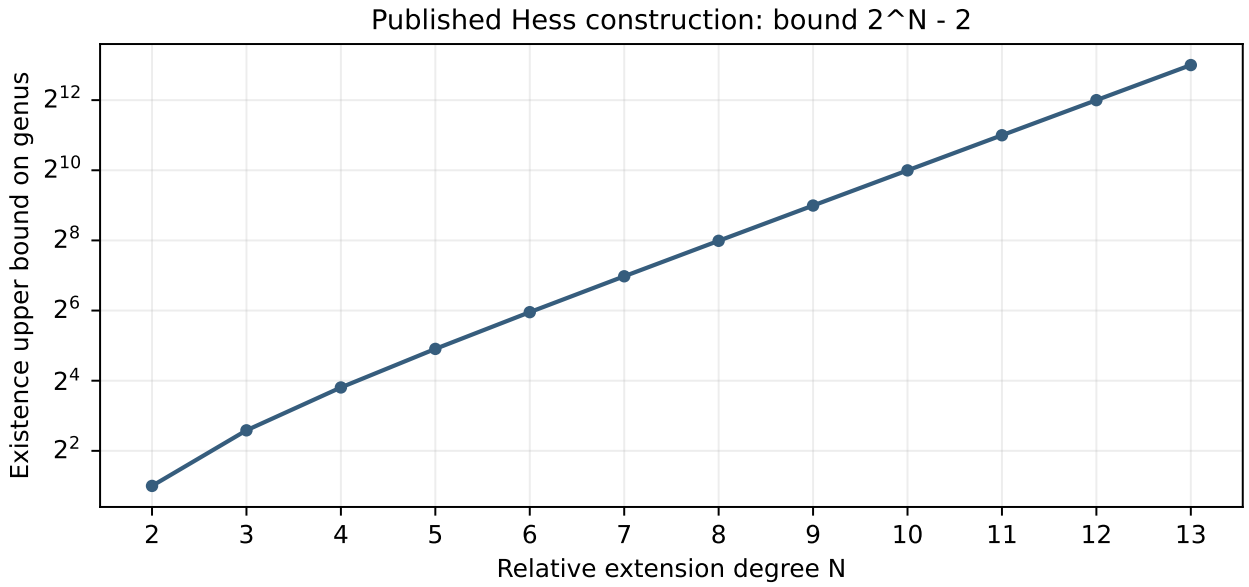


Figure 2: Exact genus upper bound from the corollary, not an optimized genus or a cost measurement. Twelve integer arguments N=2 through 13; no empirical sampling uncertainty.

The plot's integer values are direct evaluations of $2^N - 2$. The renderer is `render_hess_scope.py`. These are derived arithmetic, not experimental search rows. The software-check contract fixes the unit assertions and the figure inputs before their execution.

The retained unit output and source receipt support these software and source-review statements. The corollary is not a machine-checked Lean proof and has not received an independent mathematical review in this session.

Remaining blockers and graph audit

The 355 pending exploration jobs have not been replaced or discharged by this inclusion proof. The updated repository policy requires a frozen `experiment.protocol/v1` before new campaign data. Its pointer requires `crypto-autoresearcher` on `origin/main`. At inspected commit `dcfdd16cdf7783a059ca1e6e4cc5efbf77434b7b`, the canonical checker, protocol/export schemas and skill are absent. The implementation was located in unmerged upstream PR 2464, at head `e6527aafcafa970cc93dd51b24966162aecd84e0`. It is not on the

required `origin/main`. No substitute schema is invented and no new campaign is run under a retroactive protocol. An admitted canonical location remains needed.

The checked graph sources are the cover infrastructure's source-parity table, the previous GHS audit, and the retained branch-orbit/prior-art report. Their curves, transfers and claims are unchanged. The curve registry and isogeny routes receive no new entry: this is a symbolic inclusion in a published construction, with no instantiated new curve or verified new route.

Sources and claim labels

- **Primary, read:** F. Hess, *Generalising the GHS Attack on the Elliptic Curve Discrete Logarithm Problem*, LMS J. Comput. Math. 7 (2004), pp. 174–177, equation (10), Theorems 11, 12 and 14: https://doi.org/10.1112/S1461157000000108X. The retained primary PDF and extracted text were checked; a source hash receipt accompanies this report.
- **Prior local evidence, read:** prior-art audit and Hess presentation audit. These already identify valid higher-genus presentations for the five old families.
- **Theorem/identity:** the trace lemma and inclusion corollary are derived as above; no publication-novelty claim is made for either.
- **Exact software checks:** the Rust unit log and source hashes accompany this report; they certify the stated fixed inputs, not a new family.
- **Not established:** a novel affected source family, explicit new cover, planted-log transfer, lower total cost, calibrated speedup, or a cost impossibility theorem.